A Frequently-used Notations

Table 6: Frequently used notations.

Symbol	Description
$\mathcal{D}$	Corpus of multi-vector representations for N texts.
$X^{(i)}$	Multi-vector representation of the i-th text, $\{\vec{\mathbf{x}}_1^{(i)}, \ldots, \vec{\mathbf{x}}_{n_i}^{(i)}\}$.
n_i	Number of constituent vectors in $X^{(i)}$.
N	Number of texts in $\mathcal{D}$.
$\widehat{N}$	Total number of constituent vectors, $\widehat{N} = \sum_i n_i$.
K	Number of clusters.
d	Dimension of each constituent embedding.
m	Target maximum multi-vector cardinality.
$n_{\max}$	Maximum number of constituent vectors retained after pooling.
d'	Dimension of the raw FDE produced by MUVERA.
d_f	Count-Sketch block dimension used in the cross-block FDE construction.
κ	Number of candidates for exact SNCD refinement.
ε	Convergence tolerance.
T	Number of clustering iterations.
N_u	Number of unstable texts in a non-refresh iteration.
$\widehat{N}_u$	Total number of constituent vectors among unstable texts.

B Theoretical Analyses and Proofs

B.1 Comparison of Single- and Multi-Vector Clustering

For each multi-vector X, its mean-pooled representation is $\mu(X) = \frac{1}{|X|}\sum_{\vec{\mathbf{x}}\in X}\vec{\mathbf{x}}$. For a dissimilarity function $\mathrm{dist}(\cdot)$, the complete-linkage distance between two document clusters A and B is $\mathcal{L}(A,B) = \max_{X\in A,\, Y\in B}\mathrm{dist}(X,Y)$. We compare mean-pooled single-vector representations with the original multi-vector representations under complete-linkage HAC.

Lemma B.1. *For every $d \geq 2$, there exists a corpus with two ground-truth clusters such that: (i) its ground-truth partition is not identifiable from the mean-pooled representations alone; but (ii) complete-linkage HAC under SNCD exactly recovers the partition.*

Lemma B.1 shows that documents from different clusters can have identical mean embeddings even when their constituent vectors are clearly separated.

Theorem B.2. *Let $\mathcal{P} = \{P_1, \ldots, P_K\}$ be a ground-truth partition of a multi-vector corpus. Suppose that there exist constants $0 \leq \delta < \gamma$ satisfying: (i) for every pair of distinct documents $X, Y \in P_k$, $h(X,Y) \leq \delta$ and $h(Y,X) \leq \delta$; and (ii) for every $X \in P_k$ and $Y \in P_\ell$ with $k \neq \ell$, $g(X,Y) \geq \gamma$. Then complete-linkage HAC under SNCD exactly recovers $\mathcal{P}$ when its dendrogram is cut into K clusters.*

For two multi-vectors X and Y, we define their directed coverage radius as $h(X,Y) = \max_{\vec{\mathbf{x}}\in X}\min_{\vec{\mathbf{y}}\in Y}\|\vec{\mathbf{x}}-\vec{\mathbf{y}}\|_2$, and their support separation as $g(X,Y) = \min_{\vec{\mathbf{x}}\in X,\,\vec{\mathbf{y}}\in Y}\|\vec{\mathbf{x}}-\vec{\mathbf{y}}\|_2$. Theorem B.2 gives a direct interpretation: SNCD-based clustering succeeds whenever the largest within-cluster distance δ is smaller than the smallest cross-cluster separation γ. Particularly, the construction in Lemma B.1 satisfies $\delta = 0$ and $\gamma = \sqrt{2}$, namely, Theorem B.2 guarantees exact recovery under SNCD on the same corpus whose ground-truth partition is not identifiable after mean pooling. Together, the results establish the superiority of SNCD-based multi-vector clustering over mean-pooled single-vector clustering.

B.2 Derivative Steps for Chamfer-*K*Means

The optimization function is

$$\min_{\mathcal{P},\mathcal{M}} \mathcal{J} = \sum_{k=1}^{K}\sum_{X\in P_k}\mathrm{SD}(X, M_k) \tag{17}$$

where

$$\mathrm{SD}(X,M_k) = \sum_{\vec{\mathbf{x}}_i\in X}\min_{\vec{\boldsymbol{\mu}}_j^{(k)}\in M_k}\frac{\|\vec{\mathbf{x}}_i - \vec{\boldsymbol{\mu}}_j^{(k)}\|_2^2}{|X|} + \sum_{\vec{\boldsymbol{\mu}}_j^{(k)}\in M_k}\min_{\vec{\mathbf{x}}_i\in X}\frac{\|\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{x}}_i\|_2^2}{|M_k|}.$$

This objective function is non-differentiable due to the min operator, and we resort to alternating optimization by fixing the assignments between X and M_k.

For any constituent vector $\vec{\mathbf{x}}_i$ in a multi-vector $X = \{\vec{\mathbf{x}}_1, \vec{\mathbf{x}}_2, \cdots, \vec{\mathbf{x}}_{|X|}\}$, we set $\eta(\vec{\mathbf{x}}_i, M_k)$ as a nearest vector of $\vec{\mathbf{x}}_i$ in M_k, i.e.,

$$\eta(\vec{\mathbf{x}}_i, M_k) = \arg\min_{\vec{\boldsymbol{\mu}}_j^{(k)}\in M_k}\|\vec{\mathbf{x}}_i - \vec{\boldsymbol{\mu}}_j^{(k)}\|_2 \tag{18}$$

.

For any vector $\vec{\boldsymbol{\mu}}_j^{(k)}$ in M_k, we set $\eta(\vec{\boldsymbol{\mu}}_j^{(k)}, X)$ as its nearest vector in X, i.e.,

$$\eta(\vec{\boldsymbol{\mu}}_j^{(k)}, X) = \arg\min_{\vec{\mathbf{x}}_i\in X}\|\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{x}}_i\|_2. \tag{19}$$

As such, the objective function can be rewritten as

$$\min_{\mathcal{P},\mathcal{M}}\sum_{k=1}^{K}\sum_{X\in P_k}\left(\sum_{\vec{\mathbf{x}}_i\in X}\frac{\|\vec{\mathbf{x}}_i - \eta(\vec{\mathbf{x}}_i, M_k)\|_2^2}{|X|} + \sum_{\vec{\boldsymbol{\mu}}_j^{(k)}\in M_k}\frac{\|\vec{\boldsymbol{\mu}}_j^{(k)} - \eta(\vec{\boldsymbol{\mu}}_j^{(k)}, X)\|_2^2}{|M_k|}\right). \tag{20}$$

For each multi-vector X, by taking the derivative of the optimization objective w.r.t. $\vec{\boldsymbol{\mu}}_j^{(k)}$ as zero, we have

$$\frac{\partial}{\partial\vec{\boldsymbol{\mu}}_j^{(k)}}\left(\sum_{\substack{\vec{\mathbf{x}}_i\in X,\\ \eta(\vec{\mathbf{x}}_i,M_k)=\vec{\boldsymbol{\mu}}_j^{(k)}}}\frac{\|\vec{\mathbf{x}}_i - \vec{\boldsymbol{\mu}}_j^{(k)}\|_2^2}{|X|}\right) = \sum_{\substack{\vec{\mathbf{x}}_i\in X,\\ \eta(\vec{\mathbf{x}}_i,M_k)=\vec{\boldsymbol{\mu}}_j^{(k)}}}\frac{-2(\vec{\mathbf{x}}_i - \vec{\boldsymbol{\mu}}_j^{(k)})}{|X|},$$
$$\frac{\partial}{\partial\vec{\boldsymbol{\mu}}_j^{(k)}}\left(\frac{\|\vec{\boldsymbol{\mu}}_j^{(k)} - \eta(\vec{\boldsymbol{\mu}}_j^{(k)}, X)\|_2^2}{|M_k|}\right) = \frac{2(\vec{\boldsymbol{\mu}}_j^{(k)} - \eta(\vec{\boldsymbol{\mu}}_j^{(k)}, X))}{|M_k|}, \tag{21}$$

summing these contributions over all $X \in P_k$ and setting $\frac{\partial\mathcal{J}'}{\partial\vec{\boldsymbol{\mu}}_j^{(k)}} = 0$ yields

$$\sum_{X\in P_k}\left(\sum_{\substack{\vec{\mathbf{x}}_i\in X,\\ \eta(\vec{\mathbf{x}}_i,M_k)=\vec{\boldsymbol{\mu}}_j^{(k)}}}\frac{-2(\vec{\mathbf{x}}_i - \vec{\boldsymbol{\mu}}_j^{(k)})}{|X|} + \frac{2(\vec{\boldsymbol{\mu}}_j^{(k)} - \eta(\vec{\boldsymbol{\mu}}_j^{(k)}, X))}{|M_k|}\right) = 0.$$

This leads to

$$\sum_{X\in P_k}\left(\frac{\vec{\mu}_j^{(k)}}{|M_k|}+\sum_{\vec{x}_i\in X,\eta(\vec{x}_i,M_k)=\vec{\mu}_j^{(k)}}\frac{\vec{\mu}_j^{(k)}}{|X|}\right)$$
$$=\sum_{X\in P_k}\left(\frac{\eta(\vec{\mu}_j^{(k)},X)}{|M_k|}+\sum_{\vec{x}_i\in X,\eta(\vec{x}_i,M_k)=\vec{\mu}_j^{(k)}}\frac{\vec{x}_i}{|X|}\right).$$

Then,

$$\vec{\mu}_j^{(k)}=\frac{\sum_{X\in P_k}\left(\frac{\eta(\vec{\mu}_j^{(k)},X)}{|M_k|}+\sum_{\vec{x}_i\in X,\eta(\vec{x}_i,M_k)=\vec{\mu}_j^{(k)}}\frac{\vec{x}_i}{|X|}\right)}{\sum_{X\in P_k}\left(\frac{1}{|M_k|}+\frac{|\{\eta(\vec{x}_i,M_k)=\vec{\mu}_j^{(k)}\}_{\vec{x}_i\in X}|}{|X|}\right)} \tag{22}$$

B.3 Detailed Complexity Analysis of Chamfer-KMeans+

We provide the derivations of the complexity bounds summarized in Section 5.5. The reusable cross-block FDEs of the corpus are constructed once in $O(\widehat{N}d+Nd')$ time, as analyzed in Section 5.2.

Lazy Cluster Assignment. Consider a non-refresh iteration of Algorithm 3. Constructing the cross-block FDEs of the K current centroids costs $O(K(n_{\max}d+d'))$. The stability test evaluates the exact SNCD to the previously assigned centroid for every multi-vector and scans the K cached reference distances, contributing $O(\widehat{N}n_{\max}d+NK)$ time.

Retrieving κ candidates for the N_u uncertified multi-vectors requires $O(N_uKd_f)$ surrogate-comparison time, while exact refinement over the retrieved candidates costs $O(\kappa\widehat{N}_un_{\max}d)$. Hence, a non-refresh assignment iteration costs

$$O\Big(K(n_{\max}d+d')+NK+N_uKd_f+(\widehat{N}+\kappa\widehat{N}_u)n_{\max}d\Big),$$

and evaluates exact SNCD for only $N+\kappa N_u$ multi-vector–centroid pairs. For fixed FDE dimensions and κ, this reduces to $O(NK+\widehat{N}dn_{\max})$. A full refresh instead recomputes all NK exact distances and has assignment cost $O(\widehat{N}Kdn_{\max})$.

FDE-based Greedy Seeding. Algorithm 4 performs $K-1$ seeding rounds. Each round considers at most $2\tau=O(\log K)$ candidates, with $\tau=2+\lfloor\ln K\rfloor$. Evaluating one candidate against all N multi-vectors using the cross-block FDE surrogate costs $O(Nd_f)$. After selecting a centroid, the cached nearest-centroid objective and coverage distances are updated incrementally in $O(Nd_f)$ time and therefore do not change the dominant per-round cost. Thus, excluding the reusable corpus-FDE construction, initialization costs $O(NKd_f\log K)$.

Overall Complexity. The centroid-update step is unchanged from Chamfer-KMeans and costs $O(\widehat{N}dn_{\max})$ per iteration. Using $N_u\le N$, $\widehat{N}_u\le\widehat{N}$, and $\kappa\le K$, the assignment and update costs are bounded by

$$O\Big(K\big(d'+Nd_f+\widehat{N}dn_{\max}\big)\Big)$$

per iteration. Including the one-time corpus-FDE construction and initialization yields

$$O\Big(\widehat{N}d+Nd'+NKd_f\log K+TK\big(d'+Nd_f+\widehat{N}dn_{\max}\big)\Big).$$

For fixed FDE dimensions, this retains the $O(T\widehat{N}Kdn_{\max})$ dominant worst-case term of Chamfer-KMeans, while non-refresh iterations attain the reduced assignment bound derived above.

Finally, the multi-vector corpus, cached corpus and centroid FDEs, centroids, cached reference distances, and raw-FDE workspace require

$$O\Big(\widehat{N}d+Nd_f+Kn_{\max}d+NK+d'\Big)$$

space in total, where the $O(Kd_f)$ centroid-FDE storage is absorbed by $O(Nd_f)$ since $K\le N$.

B.4 Proofs

Proof of Lemma B.1. Let $\vec{e}_1,\vec{e}_2\in\mathbb{R}^d$ be two orthogonal unit vectors, and define

$$X^{(1)}=\{\vec{e}_1,-\vec{e}_1\},\qquad X^{(2)}=\{\vec{e}_2,-\vec{e}_2\}.$$

Construct two ground-truth clusters P_1 and P_2, each containing one or more copies of $X^{(1)}$ and $X^{(2)}$, respectively. The two document types have identical mean-pooled representations:

$$\mu(X^{(1)})=\frac{\vec{e}_1-\vec{e}_1}{2}=\mathbf{0}=\frac{\vec{e}_2-\vec{e}_2}{2}=\mu(X^{(2)}).$$

Consequently, all pairwise Euclidean distances between the pooled representations are zero. This observation is therefore unchanged under any reassignment of the documents to two clusters, and hence, the ground-truth partition is not identifiable from these representations alone. As such, complete-linkage HAC operating only on the pooled representations cannot guarantee exact recovery.

We now consider SNCD. Any two documents in the same ground-truth cluster are identical and hence have zero SNCD. Moreover, for any $\vec{x}\in X^{(1)}$ and $\vec{y}\in X^{(2)}$, orthogonality gives $\|\vec{x}-\vec{y}\|_2^2=\|\vec{x}\|_2^2+\|\vec{y}\|_2^2-2\langle\vec{x},\vec{y}\rangle=2$. Therefore, each directional term of SNCD between $X^{(1)}$ and $X^{(2)}$ equals 2, and $\mathrm{SD}(X^{(1)},X^{(2)})=4$. Thus, all within-cluster distances are zero, whereas all cross-cluster distances are four. Complete-linkage HAC consequently performs all within-cluster merges before any cross-cluster merge and exactly recovers $\{P_1,P_2\}$ when the dendrogram is cut into two clusters. □

Proof of Theorem B.2. Consider two distinct documents $X,Y\in P_k$. By the definition of $h(X,Y)$, every $\vec{x}\in X$ has a constituent vector in Y within distance δ. Hence,

$$\frac{1}{|X|}\sum_{\vec{x}\in X}\min_{\vec{y}\in Y}\|\vec{x}-\vec{y}\|_2^2\le\delta^2.$$

Applying the same argument in the reverse direction gives

$$\frac{1}{|Y|}\sum_{\vec{y}\in Y}\min_{\vec{x}\in X}\|\vec{y}-\vec{x}\|_2^2\le\delta^2.$$

Accordingly, every document pair in the same cluster satisfies $\mathrm{SD}(X,Y)\le2\delta^2$.

Now consider $X\in P_k$ and $Y\in P_\ell$, where $k\neq\ell$. The condition $g(X,Y)\ge\gamma$ implies that $\|\vec{x}-\vec{y}\|_2\ge\gamma$ for every $\vec{x}\in X$ and $\vec{y}\in Y$. It follows that $\forall\vec{x}\in X$,

$$\min_{\vec{y}\in Y}\|\vec{x}-\vec{y}\|_2^2\ge\gamma^2.$$

Consequently, each directional term of SNCD is at least γ^2, and every cross-cluster document pair satisfies $\mathrm{SD}(X, Y) \geq 2\gamma^2$. Since $\delta < \gamma$, we obtain the strict separation

$$\max_{\substack{X,Y \in P_k \\ X \neq Y}} \mathrm{SD}(X, Y) \leq 2\delta^2 < 2\gamma^2 \leq \min_{\substack{X \in P_k,\, Y \in P_\ell \\ k \neq \ell}} \mathrm{SD}(X, Y). \tag{23}$$

It remains to show that this pairwise separation implies exact recovery by complete linkage in HAC. Initially, every active cluster is a singleton and is therefore contained in one ground-truth cluster. Assume inductively that every active cluster remains contained in one member of $\mathcal{P}$. The complete-linkage distance between two active clusters contained in the same P_k is at most $2\delta^2$, whereas that between active clusters contained in different members of $\mathcal{P}$ is at least $2\gamma^2$. Whenever more than K active clusters remain, at least one ground-truth cluster contains two or more active clusters. A within-cluster merge with linkage at most $2\delta^2$ is therefore available, while every cross-cluster merge has linkage at least $2\gamma^2$. Because $2\delta^2 < 2\gamma^2$, complete linkage must select a within-cluster merge. By induction, no cross-cluster merge occurs before exactly the K ground-truth clusters remain. This proves exact recovery. □

Proof of Theorem 4.1. Let $\mathcal{J}(\mathcal{P}, \mathcal{M})$ denote the objective in Eq. (5), and let $(\mathcal{P}^{(t)}, \mathcal{M}^{(t)})$ be the state after iteration t. Exact cluster assignment gives

$$\mathcal{J}(\mathcal{P}^{(t+1)}, \mathcal{M}^{(t)}) \leq \mathcal{J}(\mathcal{P}^{(t)}, \mathcal{M}^{(t)}).$$

Fix $\mathcal{P}^{(t+1)}$ and the constituent matching indices computed at $\mathcal{M}^{(t)}$. Replacing each minimum in SNCD with its selected match defines a surrogate $Q_t(\mathcal{M})$. Since each selected distance upper bounds its corresponding minimum,

$$\mathcal{J}(\mathcal{P}^{(t+1)}, \mathcal{M}) \leq Q_t(\mathcal{M}), \qquad Q_t(\mathcal{M}^{(t)}) = \mathcal{J}(\mathcal{P}^{(t+1)}, \mathcal{M}^{(t)}).$$

For each constituent of a nonempty cluster, its term in Q_t is

$$a_{kj}\|\boldsymbol{\mu}\|_2^2 - 2\langle \mathbf{b}_{kj}, \boldsymbol{\mu} \rangle + c_{kj},$$

where $a_{kj} > 0$ and $\mathbf{b}_{kj}$ are the denominator and numerator of Eq. (8), respectively. Under $\|\boldsymbol{\mu}\|_2 = 1$, Cauchy-Schwarz shows that this term is minimized by $\mathbf{b}_{kj}/\|\mathbf{b}_{kj}\|_2$, which is exactly the normalized update. If $\mathbf{b}_{kj} = \mathbf{0}$, the term is constant on the unit sphere, so retaining the previous constituent also minimizes it. Empty clusters contribute zero and are left unchanged. Thus, $Q_t(\mathcal{M}^{(t+1)}) \leq Q_t(\mathcal{M}^{(t)})$, yielding

$$\begin{aligned}\mathcal{J}(\mathcal{P}^{(t+1)}, \mathcal{M}^{(t+1)}) &\leq Q_t(\mathcal{M}^{(t+1)}) \leq Q_t(\mathcal{M}^{(t)}) \\ &= \mathcal{J}(\mathcal{P}^{(t+1)}, \mathcal{M}^{(t)}) \leq \mathcal{J}(\mathcal{P}^{(t)}, \mathcal{M}^{(t)}).\end{aligned}$$

The objective values are therefore non-increasing and bounded below by zero, and hence converge if the iterations continue indefinitely, which concludes the proof. □

Proof of Lemma 5.1. Condition on the raw encodings. For a Count-Sketch matrix $\mathbf{S}$ with d_f rows and fixed vectors $\vec{\mathbf{a}}, \vec{\mathbf{b}}$ [13],

$$\mathbb{E}\langle \mathbf{S}\vec{\mathbf{a}}, \mathbf{S}\vec{\mathbf{b}} \rangle = \langle \vec{\mathbf{a}}, \vec{\mathbf{b}} \rangle, \quad \mathrm{Var}[\langle \mathbf{S}\vec{\mathbf{a}}, \mathbf{S}\vec{\mathbf{b}} \rangle] \leq 2\|\vec{\mathbf{a}}\|_2^2\|\vec{\mathbf{b}}\|_2^2/d_f.$$

Grouping the two terms that use the same sketch in Eq. (10), applying Cauchy-Schwarz to their covariance, and using the independence of $\mathbf{S}_0$ and $\mathbf{S}_1$ gives

$$\mathrm{Var}\left[\widehat{c}(X, Y) - \langle \vec{\mathbf{f}}_s(X), \vec{\mathbf{f}}_t(Y) \rangle - \langle \vec{\mathbf{f}}_t(X), \vec{\mathbf{f}}_s(Y) \rangle\right] \leq \frac{\Lambda_{X,Y}^2}{d_f}.$$

If one sketch were reused for both blocks, the two cross orientations would coincide and the corresponding conservative bound would instead be $2\Lambda_{X,Y}^2/d_f$. Chebyshev's inequality bounds the sketch error by $\Lambda_{X,Y}/\sqrt{d_f\delta}$ with probability at least $1-\delta$. The two MUVERA errors sum to at most 2ϵ with probability at least $1 - 2p_f$. Eq. (9) multiplies their combined similarity error by two. By the union bound, the overall success probability is $1 - 2p_f - \delta$, proving the claim. □

Proof of Lemma 5.2. Write the paired reference and current centroids as $M_k^{\mathrm{ref}} = \{\vec{\boldsymbol{\mu}}_j^{(k)}\}_{j=1}^{|M_k|}$ and $M_k = \{\vec{\mathbf{u}}_j^{(k)}\}_{j=1}^{|M_k|}$, respectively, and define $b_j = 2\|\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{u}}_j^{(k)}\|_2$. For any constituent $\vec{\mathbf{x}} \in X^{(i)}$,

$$\begin{aligned}&|\|\vec{\mathbf{x}} - \vec{\boldsymbol{\mu}}_j^{(k)}\|_2^2 - \|\vec{\mathbf{x}} - \vec{\mathbf{u}}_j^{(k)}\|_2^2| \\ &\leq 2|\vec{\mathbf{x}}^\top(\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{u}}_j^{(k)})| + |\|\vec{\boldsymbol{\mu}}_j^{(k)}\|_2^2 - \|\vec{\mathbf{u}}_j^{(k)}\|_2^2| \\ &\leq 2\|\vec{\mathbf{x}}\|_2\|\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{u}}_j^{(k)}\|_2 + |\|\vec{\boldsymbol{\mu}}_j^{(k)}\|_2 - \|\vec{\mathbf{u}}_j^{(k)}\|_2| \cdot (\|\vec{\boldsymbol{\mu}}_j^{(k)}\|_2 + \|\vec{\mathbf{u}}_j^{(k)}\|_2) \\ &\leq 2\|\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{u}}_j^{(k)}\|_2 = b_j,\end{aligned}$$

since both $\vec{\boldsymbol{\mu}}_j^{(k)}$ and $\vec{\mathbf{u}}_j^{(k)}$ are L_2-normalized.

Denote by $\mathrm{SD}_{\rightarrow}(X^{(i)}, X^{(j)})$ and $\mathrm{SD}_{\leftarrow}(X^{(i)}, X^{(j)})$ the forward and backward terms of the SNCD $\mathrm{SD}(X^{(i)}, X^{(j)})$, respectively. The elementary inequality $|\min_j a_j - \min_j c_j| \leq \max_j |a_j - c_j|$ gives

$$\left|\min_j \|\vec{\mathbf{x}} - \vec{\boldsymbol{\mu}}_j^{(k)}\|_2^2 - \min_j \|\vec{\mathbf{x}} - \vec{\mathbf{u}}_j^{(k)}\|_2^2\right| \leq \max_j b_j. \tag{24}$$

Averaging Eq. (24) over all $\vec{\mathbf{x}} \in X^{(i)}$ preserves this bound for the forward term of SNCD, i.e., $|\mathrm{SD}_{\rightarrow}(X^{(i)}, M_k^{\mathrm{ref}}) - \mathrm{SD}_{\rightarrow}(X^{(i)}, M_k)| \leq \max_j b_j$. For the backward term, applying the same inequality for every paired centroid constituent gives

$$\left|\min_{\vec{\mathbf{x}} \in X^{(i)}} \|\vec{\boldsymbol{\mu}}_j^{(k)} - \vec{\mathbf{x}}\|_2^2 - \min_{\vec{\mathbf{x}} \in X^{(i)}} \|\vec{\mathbf{u}}_j^{(k)} - \vec{\mathbf{x}}\|_2^2\right| \leq b_j. \tag{25}$$

Similarly, averaging Eq. (25) over the centroid constituents leads to $|\mathrm{SD}_{\leftarrow}(X^{(i)}, M_k^{\mathrm{ref}}) - \mathrm{SD}_{\leftarrow}(X^{(i)}, M_k)| \leq \frac{1}{|M_k|}\sum_j b_j$. Then, adding the two directed bounds yields B_k in Eq. (12) and hence

$$\left|\mathrm{SD}(X^{(i)}, M_k) - \mathrm{SD}(X^{(i)}, M_k^{\mathrm{ref}})\right| \leq B_k. \tag{26}$$

For every alternative centroid M_k with $k \neq k_i$, Eq. (26) implies

$$\mathrm{SD}(X^{(i)}, M_k) \geq \mathrm{SD}(X^{(i)}, M_k^{\mathrm{ref}}) - B_k.$$

Combining this lower bound with Eq. (13) yields $\mathrm{SD}(X^{(i)}, M_{k_i}) \leq \mathrm{SD}(X^{(i)}, M_k)$ for every $k \neq k_i$. Therefore, M_{k_i} minimizes the current exact SNCD for $X^{(i)}$, proving the lemma. □

Table 7: Multi-vector representation configurations for the evaluated datasets.

Dataset	Encoder	Pooling	Target m
TREC	Jina	ToMe	12
Reuters-52	SBERT	ToMe	24
20 Newsgroups	SBERT	ToMe	32
SciDocs-MAG	SBERT	ToMe	12
SciDocs-MESH	SBERT	ToMe	12
LoCoV1	Jina	ToMe	32
DBpedia	SBERT	ToMe	32
Yahoo Answers	SBERT	ToMe	12

C Raw FDE Construction

Algorithm 5 gives the partition-and-aggregation stage of the FDE construction adapted from Dhulipala et al. [20]. Let R, P, and d_p denote the number of repetitions, the number of SimHash bits, and the inner-projection dimension, respectively; throughout the experiments, we use $R = 8$, $P = 4$ (thus $B = 2^P = 16$), and $d_p = 32$, with all random maps fixed globally. We fix R SimHash maps $\phi_r : \mathbb{R}^d \to \{0, \dots, B-1\}$ and inner projections $\psi_r : \mathbb{R}^d \to \mathbb{R}^{d_p}$; our implementation instantiates ψ_r with a sparse AMS sketch [2]. The same maps are shared by all multi-vectors and centroids, and we omit them from the algorithm input for brevity.

For each repetition and bucket, the source role stores the sum of its projected vectors, whereas the target role stores their average, corresponding to the query and document roles in the original MUVERA construction. An empty source-role bucket remains zero, whereas an empty target-role bucket is filled by the projected vector whose P-bit SimHash code is closest to that bucket in Hamming distance, with ties broken arbitrarily. Concatenating the $R \times B$ blocks gives raw encodings in $\mathbb{R}^{d'}$, where $d' = R \times B \times d_p$. Algorithm 5 returns the normalized raw source and target encodings $\vec{\mathbf{f}}_s(X)$ and $\vec{\mathbf{f}}_t(X)$. Scaling both encodings by $R^{-1/2}$ averages their inner product over the R repetitions, while the additional factor $|X|^{-1}$ in the source encoding yields the normalized Chamfer similarity.

Algorithm 5: The MUVERA Algorithm [20]

Input: Multi-vector X, parameters R, P, d_p.
Output: Raw encodings $\vec{\mathbf{f}}_s(X), \vec{\mathbf{f}}_t(X) \in \mathbb{R}^{d'}$.

for $r \leftarrow 1$ **to** R **do**
  for $b \leftarrow 0$ **to** $B-1$ **do**
    $X_{r,b} \leftarrow \{\vec{\mathbf{x}} \in X : \phi_r(\vec{\mathbf{x}}) = b\}$;
    $Q_{r,b} \leftarrow \sum_{\vec{\mathbf{x}} \in X_{r,b}} \psi_r(\vec{\mathbf{x}})$;
    if $X_{r,b} \neq \emptyset$ **then**
      $D_{r,b} \leftarrow |X_{r,b}|^{-1} \sum_{\vec{\mathbf{x}} \in X_{r,b}} \psi_r(\vec{\mathbf{x}})$;
    else
      $\vec{\mathbf{x}}^* \leftarrow \arg\min_{\vec{\mathbf{x}} \in X} \mathrm{Ham}(\phi_r(\vec{\mathbf{x}}), b)$;
      $D_{r,b} \leftarrow \psi_r(\vec{\mathbf{x}}^*)$;
$Q_X \leftarrow R^{-1/2} \|_{r,b} Q_{r,b}$; $\quad D_X \leftarrow R^{-1/2} \|_{r,b} D_{r,b}$;
$\vec{\mathbf{f}}_s(X) \leftarrow Q_X / |X|$; $\quad \vec{\mathbf{f}}_t(X) \leftarrow D_X$;
return $\vec{\mathbf{f}}_s(X), \vec{\mathbf{f}}_t(X)$;

D Detailed Experimental Settings

D.1 Datasets

We provide additional details on the datasets used in our experiments. When predefined training and test splits are available, we combine them to form the clustering corpus unless otherwise specified. Ground-truth labels are used only for clustering evaluation.

TREC. TREC is a question-classification benchmark introduced by Li and Roth [43], where each question is assigned to a semantic category according to the type of answer it seeks. We use the preprocessed version distributed in the Text-Classification dataset collection[1]. We combine the provided training and test portions into a single clustering corpus. The resulting dataset contains 5,952 questions and 47 ground-truth classes.

Reuters-52* and *20 Newsgroups. We use the preprocessed versions distributed by Hendrycks and Gimpel [34][2]. *Reuters-52*, derived from Reuters-21578 [42], contains 9,100 news texts from 52 categories after combining its training and test splits. *20 Newsgroups* [41] contains 18,211 texts from 20 Usenet discussion groups.

SciDocs-MAG* and *SciDocs-MESH. We use the MAG and MESH document-classification tasks from the official SciDocs release [15][3]. *SciDocs-MAG* contains 25,003 scientific papers from 19 research categories, while *SciDocs-MESH* contains 23,481 biomedical papers from 11 MeSH disease categories.

LoCoV1. We use the document collection from the official LoCoV1-Documents release [57][4]. It contains 14,838 documents from 12 constituent datasets. For clustering, we use the text in the `passage` field and treat the originating `dataset` identifier as the ground-truth class.

DBpedia* and *Yahoo Answers. We use the ontology- and topic-classification datasets introduced by Zhang et al. [76] from the Text-Classification collection[5]. Combining the predefined training and test splits yields 630,000 texts from 14 classes for *DBpedia* and 1.46 million texts from 10 classes for *Yahoo Answers*.

D.2 Multi-vector Representation Construction

Encoders and pooling. Table 7 outlines the representation configuration. We use `sentence-transformers/all-mpnet-base-v2` for SBERT and `jinaai/jina-embeddings-v4` for Jina; for the latter, we use the `text-matching` task configuration. Multi-vector representations are constructed from the constituent embeddings produced by these encoders and subsequently reduced using bipartite token merging (ToMe) [8]. Dataset-specific encoder choices and target cardinalities are summarized in Table 7 and used as defaults unless otherwise specified. Documents containing fewer than m constituent vectors are not artificially expanded during pooling and

[1] https://github.com/Renovamen/Text-Classification/blob/master/notes/datasets.md
[2] https://github.com/hendrycks/error-detection/tree/master/NLP/Categorization
[3] https://github.com/allenai/scidocs
[4] https://huggingface.co/datasets/hazyresearch/LoCoV1-Documents
[5] https://github.com/Renovamen/Text-Classification/blob/master/notes/datasets.md

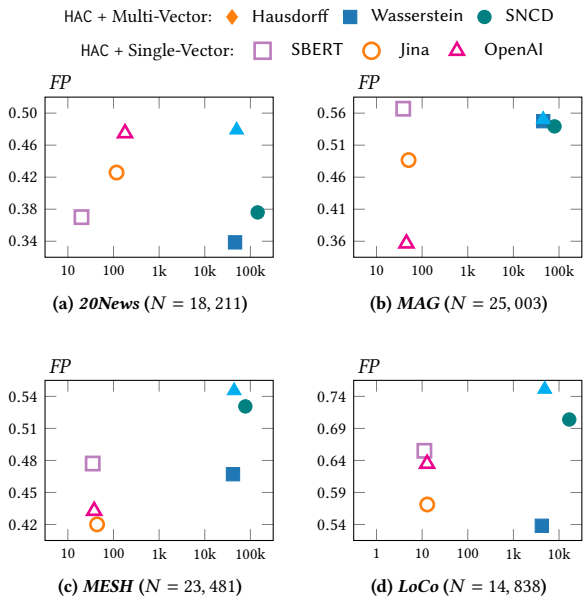


Figure 11: Clustering quality vs. CPU runtime with HAC.

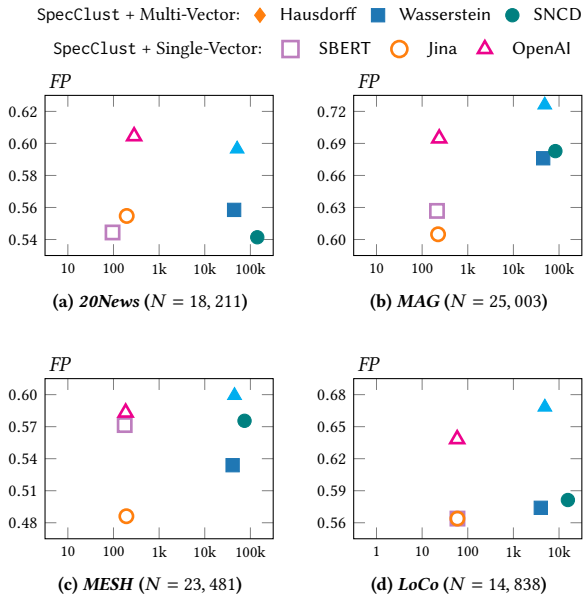


Figure 12: Clustering quality vs. CPU runtime with SpecClust.

therefore retain smaller cardinalities. Additional analyses of pooling strategies, encoder choices, and target multi-vector cardinality are reported in Appendix E.1.

D.3 Baseline Details

Single-vector baselines. For single-vector clustering, each document is represented using SBERT, Jina, or OpenAI. For OpenAI, we use `text-embedding-3-small`. For Jina, we use the `text-matching` task configuration. Each representation is combined with K-Means, HAC, or SpecClust. The K-Means baseline uses k-means++ initialization, HAC uses complete linkage, and SpecClust operates on a precomputed Gaussian affinity matrix constructed from normalized pairwise distances.

Generic multi-vector baselines. We consider fixed-dimensional FDE representations constructed with MUVERA [20] and combine them with K-Means, HAC, or SpecClust. We also directly compute pairwise Wasserstein, Hausdorff, and SNCD distances between multi-vectors and combine them with HAC or SpecClust. The resulting SNCD-based baselines are denoted by HAC+SNCD and SpecClust+SNCD.

ClusterLLM. We evaluate ClusterLLM [77], the granularity-selection pipeline using OpenAI `GPT-4o-Mini`. The method uses zero-shot pairwise Yes/No queries with temperature 0 and no in-context demonstrations. Its pair-query budget is $\max(\lfloor 0.1N \rfloor, K)$. The procedure uses HAC over precomputed embeddings.

k-LLMmeans. We evaluate k-LLMmeans [21] using `GPT-4o-Mini`. The method is initialized with k-means++ and performs three outer LLM-assisted iterations. At each iteration, one summary is generated for each cluster from at most 20 representative documents, after which the summaries are embedded locally using the same dataset-specific encoder as the document representations. Summary generation uses temperature 0, and the internal K-Means optimization uses convergence tolerance 10^{-4}.

D.4 Evaluation Metrics

We evaluate clustering quality using normalized mutual information (NMI), FP, and clustering accuracy (ACC), with FP as the primary metric. All three metrics take values in $[0, 1]$, and larger values indicate better clustering quality.

Let $y_i \in \{1, \dots, K\}$ denote the ground-truth class label of the i-th text, represented by $X^{(i)}$, and let $\hat{y}_i \in \{1, \dots, K\}$ denote its predicted cluster label. We define the contingency count

$$n_{kc} = |\{i : \hat{y}_i = k,\ y_i = c\}|, \tag{27}$$

and let $n_{k\cdot} = \sum_c n_{kc}$, $n_{\cdot c} = \sum_k n_{kc}$, and $N = \sum_{k,c} n_{kc}$.

FP. FP is the harmonic mean of Purity and Inverse Purity [25, 64]. Purity measures the extent to which each predicted cluster is dominated by a single ground-truth class:

$$\text{Purity} = \frac{1}{N} \sum_{k=1}^{K} \max_c n_{kc}. \tag{28}$$

Conversely, Inverse Purity measures the extent to which each ground-truth class is concentrated in a single predicted cluster:

$$\text{InvPurity} = \frac{1}{N} \sum_{c=1}^{K} \max_k n_{kc}. \tag{29}$$

Their harmonic mean is

$$\text{FP} = \frac{2\,\text{Purity}\,\text{InvPurity}}{\text{Purity} + \text{InvPurity}}. \tag{30}$$

We define FP $= 0$ when the denominator is zero. FP therefore captures both cluster purity and class completeness.

NMI. Let

$$p_{kc} = \frac{n_{kc}}{N}, \qquad p_k = \frac{n_{k\cdot}}{N}, \qquad q_c = \frac{n_{\cdot c}}{N}. \tag{31}$$

Table 8: Effect of different pooling strategies on multi-vector clustering.

Dataset	Encoder	m	Pooling	NMI	FP	ACC
TREC	Jina	12	ToMe	**0.4748**	**0.5395**	**0.4106**
			Hierarchical	0.4217	0.4654	0.3511
			K-Means	0.4201	0.4794	0.4022
			Sequential	0.4054	0.4777	0.3560
R-52	SBERT	24	ToMe	**0.5673**	**0.4928**	**0.3056**
			Hierarchical	0.5387	0.4609	0.2870
			K-Means	0.5291	0.4396	0.2491
			Sequential	0.5152	0.3886	0.2051
20News	SBERT	32	ToMe	**0.6040**	**0.6534**	**0.6351**
			Hierarchical	0.5935	0.6459	0.6203
			K-Means	0.5909	0.6388	0.5860
			Sequential	0.5878	0.6064	0.5814
MAG	SBERT	12	ToMe	**0.7117**	**0.7586**	**0.7171**
			Hierarchical	0.7066	0.7489	0.7110
			K-Means	0.6950	0.7284	0.6814
			Sequential	0.6802	0.6924	0.6335
MESH	SBERT	12	ToMe	**0.5271**	0.6817	0.6045
			Hierarchical	0.5206	**0.6833**	**0.6093**
			K-Means	0.5200	0.6824	0.6076
			Sequential	0.4785	0.6017	0.5500
LoCo	Jina	32	ToMe	**0.6635**	**0.7541**	**0.5333**
			Hierarchical	OOT	OOT	OOT
			K-Means	0.6553	0.6661	0.4902
			Sequential	0.6381	0.6005	0.4251

The mutual information between the predicted clustering and the ground-truth partition is

$$I(\hat{Y};Y) = \sum_{k,c:\, n_{kc}>0} p_{kc} \log \frac{p_{kc}}{p_k q_c}, \tag{32}$$

and their entropies are

$$H(\hat{Y}) = -\sum_{k:\, p_k>0} p_k \log p_k, \qquad H(Y) = -\sum_{c:\, q_c>0} q_c \log q_c. \tag{33}$$

Following the arithmetic normalization used by our implementation, NMI is defined as

$$\mathrm{NMI} = \frac{2I(\hat{Y};Y)}{H(\hat{Y}) + H(Y)}. \tag{34}$$

NMI measures the normalized mutual dependence between the predicted and ground-truth partitions.

ACC. Since cluster identifiers have no intrinsic correspondence to ground-truth class identifiers, ACC is computed after an optimal one-to-one matching between predicted clusters and ground-truth classes. Let $\mathfrak{S}_K$ denote the set of permutations of $\{1,\dots,K\}$. We define

$$\mathrm{ACC} = \max_{\pi\in\mathfrak{S}_K} \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[y_i = \pi(\hat{y}_i)]\,. \tag{35}$$

We compute the optimal mapping π using the Hungarian algorithm.

D.5 Implementation Details

Following standard text-clustering evaluation, we set the number of clusters K to the number of ground-truth classes for each dataset.

Algorithm Settings. Both `Chamfer-KMeans` and `Chamfer-KMeans+` run for at most 500 iterations with convergence tolerance $\varepsilon = 10^{-4}$. For the Count-Sketch block dimension, we use $d_f = 256$ for *TREC*, *SciDocs-MAG*, *LoCoV1*, and *Yahoo Answers*, and $d_f = 128$ for *Reuters-52*, *20 Newsgroups*, *SciDocs-MESH*, and *DBpedia*. Unless otherwise specified, `Chamfer-KMeans+` retrieves $\kappa = 4$ candidate centroids before exact SNCD refinement and uses the assignment-refresh threshold $\rho = 0.3$.

Centroid Cardinality. ToMe caps each multi-vector at m constituent vectors; representations with fewer than m vectors remain shorter and hence retain variable cardinalities after pooling.

For efficient batched centroid computation, `Chamfer-KMeans` and `Chamfer-KMeans+` use the fixed centroid cardinality

$$n_{\max} = \max_i |X^{(i)}|, \qquad n_{\max} \le m.$$

Whenever a multi-vector is used to initialize a centroid, one with fewer than $n_{\max}$ constituent vectors is padded with its mean constituent representation, followed by row-wise L_2 normalization. This operation is used during initialization and, when necessary, empty-cluster reinitialization; it is not applied to the multi-vector representations used for SNCD computation.

For batched computation, variable-length multi-vectors are temporarily zero-padded with masked rows, which do not contribute to SNCD evaluation or centroid updates.

Runtime Measurement. Runtime is measured in wall-clock time, excluding dataset loading, cluster output, embedding construction, and ToMe pooling.

Hardware. CPU experiments are conducted on a Linux node with two Intel Xeon Gold 6330 CPUs (2.0 GHz; 56 physical cores in total) and 1 TiB RAM. GPU experiments use a single NVIDIA A100 GPU (80 GB) on a separate Linux node with AMD EPYC 9754 CPUs and 1.5 TiB RAM.

E Additional Experimental Results

E.1 Representation Design Choices

Unless otherwise specified, the multi-vector analyses in this subsection use `Chamfer-KMeans`, so the corresponding default-configuration results coincide with those reported in the main clustering-quality evaluation.

Single-vector versus multi-vector clustering. Results on the remaining four main datasets (Figures 11 and 12) confirm the same effectiveness–efficiency tradeoff as in Section 3.1: multi-vector clustering with SNCD generally improves clustering quality at substantially higher computational cost.

Pooling Strategies. We compare ToMe with hierarchical, K-`Means`, and sequential pooling [14, 54], using the `PyLate`[6] implementations for the three baselines. For each dataset, we fix the encoder and target cardinality m, vary only the pooling strategy, and cluster with `Chamfer-KMeans`.

[6] https://github.com/stefan-jo/pylate

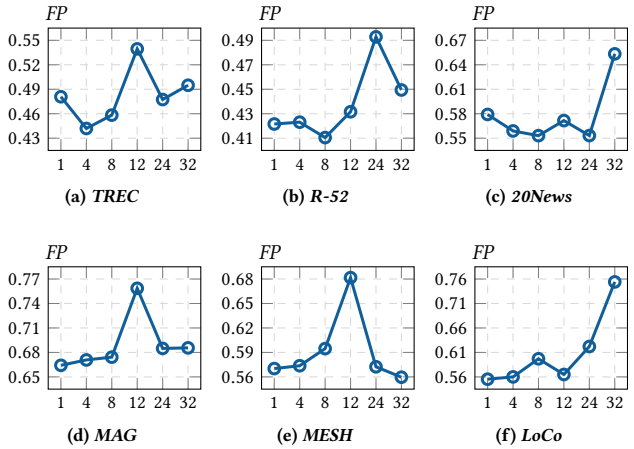


Figure 13: Effect of varying the target multi-vector size m on clustering performance.

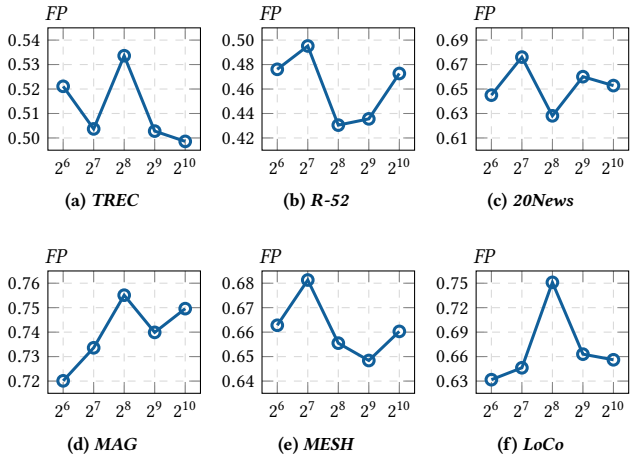


Figure 14: Effect of varying the Count-Sketch block dimension d_f on clustering performance.

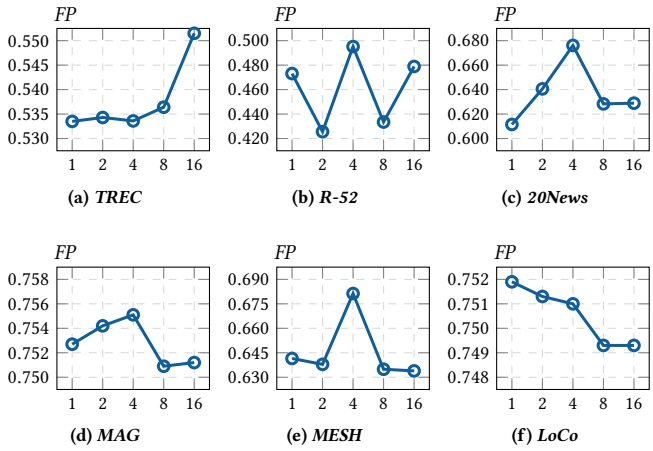


Figure 15: Effect of varying the candidate-set size κ on clustering performance.

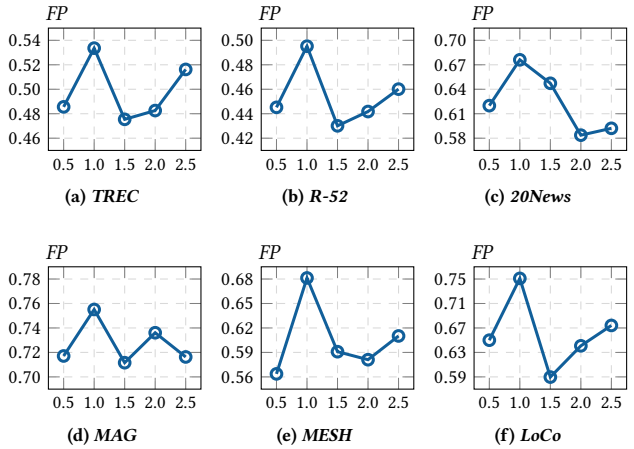


Figure 16: Effect of scaling the local-trial budget relative to the default $\tau = 2 + \lfloor \ln K \rfloor$ on clustering performance. The x-axis reports the budget multiplier.

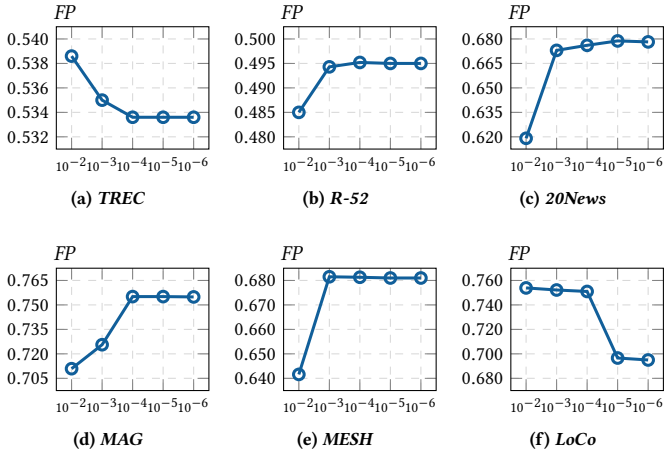


Figure 17: Effect of varying the convergence tolerance ε on clustering performance.

Table 9: Effect of different encoder representations on multi-vector and single-vector clustering.

Dataset	Encoder	Representation	Pooling	NMI	FP	ACC
TREC	Jina	Multi-vector ($m = 12$)	ToMe	**0.4748**	**0.5395**	**0.4106**
		Single-vector	–	0.4447	0.4500	0.3592
	SBERT	Multi-vector ($m = 12$)	ToMe	0.3845	0.3547	0.2572
		Single-vector	–	0.3484	0.3384	0.2503
LoCo	Jina	Multi-vector ($m = 32$)	ToMe	**0.6635**	**0.7541**	**0.5333**
		Single-vector	–	0.5970	0.6522	0.3354
	SBERT	Multi-vector ($m = 32$)	ToMe	0.6499	0.6570	0.3782
		Single-vector	–	0.5974	0.5761	0.3764
DBpedia	Jina	Multi-vector ($m = 32$)	ToMe	0.6144	0.6558	0.6199
		Single-vector	–	0.5306	0.5735	0.4682
	SBERT	Multi-vector ($m = 32$)	ToMe	**0.8630**	**0.8916**	**0.8526**
		Single-vector	–	0.7681	0.7667	0.6693

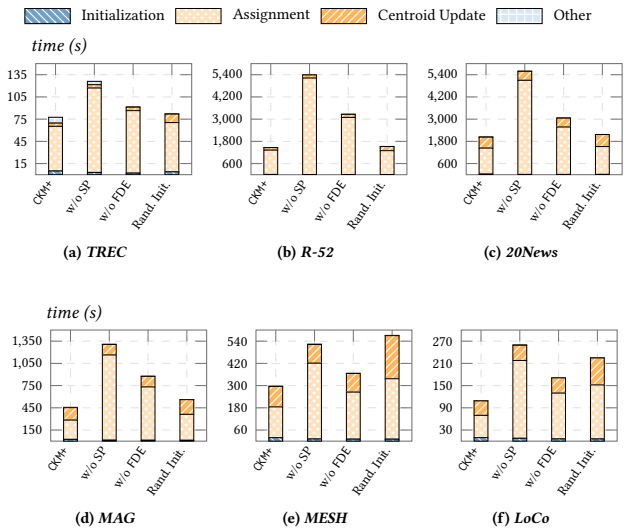


Figure 18: CPU runtime breakdown of `Chamfer-KMeans+` and its ablation variants on the six main datasets, decomposed into initialization, cluster assignment, centroid update, and other overhead. `w/o SP`, `w/o FDE`, and `Rand. Init.` denote removing stable-text pruning, removing FDE candidate retrieval, and using random initialization, respectively.

Table 8 shows ToMe achieves the highest NMI throughout among the five datasets completed by all four strategies; hierarchical pooling gives slightly higher FP and ACC on *SciDocs-MESH* but is OOT on *LoCoV1*. We therefore use ToMe as the default pooling strategy.

Encoder Choices. Table 9 compares each encoder under multi-vector clustering with `Chamfer-KMeans` and single-vector clustering with `K-Means`. Across *TREC*, *LoCoV1*, and *DBpedia*, the multi-vector configuration improves NMI, FP, and ACC over its single-vector counterpart under each encoder. The relative ranking of the encoders is dataset-dependent: Jina performs better on *TREC* and *LoCoV1*, whereas SBERT is substantially stronger on *DBpedia*. These results further show that both representation form and encoder choice affect clustering quality.

Target Multi-Vector Cardinality. We vary the target cardinality $m \in \{1, 4, 8, 12, 24, 32\}$ using `Chamfer-KMeans` to isolate representation granularity from the optimizations in `Chamfer-KMeans+`. Figure 13 shows dataset-dependent optima: $m = 12$ for *TREC*, *SciDocs-MAG*, and *SciDocs-MESH*, $m = 24$ for *Reuters-52*, and $m = 32$ for *20 Newsgroups* and *LoCoV1*. Hence, increasing m does not uniformly improve clustering quality.

E.2 Initialization Design Choices

Table 10 reports the absolute clustering quality corresponding to the relative differences shown in Figure 9. For candidate generation, we compare random initialization, two objective-guided-only variants using τ and 2τ local trials, respectively, and our full objective- and coverage-guided strategy. For coverage aggregation, we keep the candidate-generation mechanism fixed and compare the minimum, harmonic, geometric, and arithmetic rules.

For $a, b \geq 0$, let $a = \widehat{d}_{\rightarrow}(X, Y)$ and $b = \widehat{d}_{\leftarrow}(X, Y)$. The four coverage-aggregation rules considered above satisfy

$$2\min\{a, b\} \leq \frac{4ab}{a+b} \leq 2\sqrt{ab} \leq a + b, \tag{36}$$

corresponding respectively to the minimum, harmonic, geometric, and arithmetic rules. The minimum rule is therefore the most conservative aggregation: it assigns a large coverage score only when both directional surrogate distances are large.

E.3 Detailed Ablation Results

The per-iteration results in Table 11 isolate assignment cost from changes in convergence behavior. Relative to `Chamfer-KMeans+`, removing stable-text pruning and FDE-based candidate retrieval increases the average assignment time per iteration by 1.94–5.25× and 1.44–3.29×, respectively.

E.4 Complete Parameter Analysis

We complement Section 6.5 with full sensitivity results on all six main datasets, varying d_f, κ, τ, and ε one at a time while holding the remaining settings fixed.

Count-Sketch block dimension. Figure 14 shows that the effect of d_f is non-monotonic and dataset-dependent. The main experiments use $d_f = 256$ for *TREC*, *SciDocs-MAG*, and *LoCoV1*, and $d_f = 128$ for *Reuters-52*, *20 Newsgroups*, and *SciDocs-MESH*. These results also show that increasing the block dimension does not universally improve clustering quality.

Candidate-set size. Figure 15 varies the number κ of candidate centroids retrieved before exact SNCD refinement. The default setting $\kappa = 4$ achieves the highest FP on *Reuters-52*, *20 Newsgroups*, *SciDocs-MAG*, and *SciDocs-MESH*. On *LoCoV1*, the results are nearly unchanged for $\kappa \in \{1, 2, 4\}$, whereas *TREC* benefits from a larger candidate set. We therefore use $\kappa = 4$ as a common default that provides strong clustering quality while keeping the exact refinement set small.

Local-trial budget. Figure 16 scales the local-trial budget from 0.5τ to 2.5τ, where $\tau = 2 + \lfloor \ln K \rfloor$ is the default. The default setting achieves the highest FP on all six datasets, while increasing the budget does not consistently improve clustering quality. This supports the use of the lightweight default budget.

Convergence tolerance. Figure 17 shows no consistent benefit from tightening ε below 10^{-4}: quality is largely unchanged, improves only slightly on *20 Newsgroups*, and decreases on *LoCoV1*. We therefore use $\varepsilon = 10^{-4}$ throughout.

E.5 Additional Runtime Analysis

Figure 18 provides a CPU runtime breakdown for `Chamfer-KMeans+` and the three component-ablation variants considered in Figure 8. We decompose the end-to-end runtime into initialization, cluster assignment, centroid update, and other overhead.

For `Chamfer-KMeans+`, cluster assignment accounts for 55.0–89.6% of the end-to-end CPU runtime across the six datasets. This fraction increases to 78.1–96.8% without stable-text pruning and to 69.3–94.7% without FDE-based candidate retrieval, confirming that both mechanisms primarily reduce the dominant assignment cost.

Table 10: Absolute clustering-quality results for the initialization design choices evaluated in Figure 9.

Ablation	Variant	*TREC*			*R-52*			*20News*			*MAG*			*MESH*			*LoCo*		
		NMI	FP	ACC	NMI	FP	ACC	NMI	FP	ACC	NMI	FP	ACC	NMI	FP	ACC	NMI	FP	ACC
Candidate Generation	Random Initialization	0.4310	0.4782	0.3841	0.5333	0.4474	0.2526	0.5418	0.5555	0.4831	0.6806	0.6892	0.5946	0.3924	0.5278	0.4356	0.6152	0.6063	0.4445
	Objective-only (τ)	0.4566	0.4977	0.4024	0.5409	0.4179	0.2445	0.5684	0.5889	0.5382	0.6875	0.7105	0.6702	0.4585	0.5635	0.4642	0.6199	0.6209	0.4650
	Objective-only (2τ)	0.4720	0.5126	0.4015	0.5472	0.4582	0.2735	0.5758	0.6084	0.5915	0.6756	0.6954	0.6314	0.4610	0.5824	0.5230	0.6247	0.6155	0.4475
	Objective + Coverage	**0.4931**	**0.5336**	**0.4152**	**0.5579**	**0.4952**	**0.3111**	**0.6179**	**0.6761**	**0.6307**	**0.7060**	**0.7551**	**0.7252**	**0.5338**	**0.6813**	**0.6110**	**0.6779**	**0.7510**	**0.5228**
Coverage Combination	**Minimum**	**0.4931**	**0.5336**	**0.4152**	**0.5579**	**0.4952**	**0.3111**	**0.6179**	**0.6761**	**0.6307**	**0.7060**	**0.7551**	**0.7252**	**0.5338**	**0.6813**	**0.6110**	**0.6779**	**0.7510**	0.5228
	Harmonic	0.4805	0.4851	0.3648	0.5498	0.4622	0.2749	0.5877	0.6281	0.5938	0.6898	0.7193	0.6747	0.4965	0.6091	0.5444	0.6499	0.6480	0.5152
	Geometric	0.4754	0.5025	0.3918	0.5432	0.4650	0.2735	0.5808	0.6070	0.5624	0.6856	0.7140	0.6442	0.4875	0.6098	0.5366	0.6467	0.6670	**0.5389**
	Arithmetic	0.4637	0.4913	0.3817	0.5371	0.4262	0.2527	0.5763	0.6202	0.5822	0.6793	0.7058	0.6518	0.4843	0.6135	0.5724	0.6265	0.6427	0.4852

Table 11: Detailed component-ablation results of `Chamfer-KMeans+` on the six main datasets. Besides NMI, FP, and ACC, we report convergence iterations and the average per-iteration CPU time for cluster assignment and centroid update. For reference, the iteration count of `Chamfer-KMeans` is also included.

Dataset	Variant	NMI	FP	ACC	Iterations		Per-Iteration CPU Time (s)	
					Variant	`Chamfer-KMeans`	Assign.	Update
TREC	**`Chamfer-KMeans+`**	0.4931	0.5336	0.4152	36	47	1.6751	0.1225
	No Stable-Text Pruning	0.4909	0.5369	0.4116	35		3.2571	0.1301
	No FDE Candidate Retrieval	0.4909	0.5369	0.4116	35		2.4115	0.1336
	Random Initialization	0.4310	0.4782	0.3841	42		1.5833	0.2723
R-52	**`Chamfer-KMeans+`**	0.5579	0.4952	0.3111	79	98	16.6318	1.7511
	No Stable-Text Pruning	0.5526	0.4902	0.3078	86		60.6225	1.9498
	No FDE Candidate Retrieval	0.5526	0.4902	0.3078	86		35.8896	1.9459
	Random Initialization	0.5333	0.4474	0.2526	102		12.6669	2.2067
20News	**`Chamfer-KMeans+`**	0.6179	0.6761	0.6307	114	119	12.3210	5.1838
	No Stable-Text Pruning	0.6088	0.6648	0.6197	89		57.1126	5.4490
	No FDE Candidate Retrieval	0.6088	0.6648	0.6197	89		28.7737	5.4036
	Random Initialization	0.5418	0.5555	0.4831	98		15.3453	6.6337
MAG	**`Chamfer-KMeans+`**	0.7060	0.7551	0.7252	70	58	3.7170	2.4042
	No Stable-Text Pruning	0.7035	0.7505	0.7217	59		19.5052	2.4106
	No FDE Candidate Retrieval	0.7035	0.7505	0.7217	59		12.2124	2.4125
	Random Initialization	0.6806	0.6892	0.5946	82		4.2775	2.4192
MESH	**`Chamfer-KMeans+`**	0.5338	0.6813	0.6110	49	77	3.3925	2.2585
	No Stable-Text Pruning	0.5374	0.6826	0.6100	44		9.3034	2.2912
	No FDE Candidate Retrieval	0.5374	0.6826	0.6100	44		5.7727	2.2875
	Random Initialization	0.3924	0.5278	0.4356	102		3.2016	2.2821
LoCo	**`Chamfer-KMeans+`**	0.6779	0.7510	0.5228	45	33	1.3342	0.8707
	No Stable-Text Pruning	0.6759	0.7491	0.5200	46		4.5686	0.8937
	No FDE Candidate Retrieval	0.6759	0.7491	0.5200	46		2.6882	0.8949
	Random Initialization	0.6152	0.6063	0.4445	81		1.8004	0.9009